Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 1 of 5 · Question 2 part d · 5 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

2 (a)	• $\frac{6.1}{100}$; • x85; • 5.2;	Allow ecf for mps 2 and 3 If 5.18 given for mp3 then max 2 marks Full marks for correct final answer	3
(b)	• add sample to boiling tube; • add biuret solution; • purple if present, (remains blue if absent);		3
(c)	• $\frac{100}{16}$; • = 6.25/7;	ecf 1 mark max for correct final answer to a division calculation	2
(d)	Any five from: • contains calcium; • contains vitamin D; • vitamin D allows calcium uptake/absorption (from intestine); • calcium absorbed/stored in bones/used to form compact bone; • to strengthen bones/increases bone density; • also contains protein which is required for (bone) development/growth;	Allow makes bones less brittle	5

Total 13

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 1 of 5 · Question 2 part d · 5 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 2 of 5 · Question 2 part c · 6 marks

2019 JUN · Paper 02 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

2(a)	A; (1 and 3) (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(i)	D spongy bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(ii)	B compact bone; (1)		**(1)**

Question Number	Answer	Additional guidance	Mark
2(b)(iii)	cartilage;		**(1)**

Question Number	Answer	Additional guidance	**Mark**
2(b)(iv)	shock absorber/cushions; allows free movement of joint/prevents grinding/reduces friction;		**(2)**

Question Number	Answer	Additional guidance	Mark
2(c)	• Ca^{2+}needed; • forms part of bone structure / provides strength to bone/no calcium bones are less dense/weaker; • protein required; • for muscle development/component • of muscle; • vitamin D needed; • to allow uptake of Ca^{2+};		**(6)**

Total 12 marks

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 2 of 5 · Question 2 part c · 6 marks

2019 JUN · Paper 02 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Paper 2: Subject Human Biology

Question Number	Answer	Additional guidance	Mark

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 3 of 5 · Question 2 part b · 3 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

number				
2 (a)	(i)	1 = humerus; 2 = radius; 3 = ulna; 4 = fingers/phalanges;		4
	(ii)	circle at the elbow to include all three bones;	accept a circle around relevant bones in the fingers	1
	(iii)	• ball and socket; • gliding/sliding;	reject named examples	2
(b)		• assists/allows uptake of calcium (salts from gut); • (calcium) needed for bone growth; • hard bone/compact bone;		3

Total 10 marks

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 3 of 5 · Question 2 part b · 3 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 4 of 5 · Question 10 part a · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number				
10	(a)	Any three from the following: • bones break more easily; • loss of height over time; • poor posture/difficulty walking/exercising; • (back) pain;		3
	(b)	Any three from the following: • muscles attached to bones; • (muscles) work antagonistically; • When one muscle contracts the other relaxes; • bone is pulled (to bring about movement);		3

Total marks 6

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 4 of 5 · Question 10 part a · 3 marks

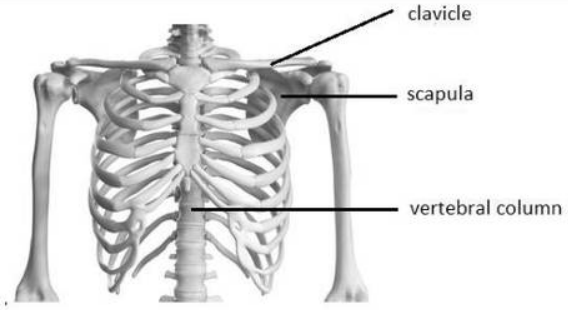
2022 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

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Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 5 of 5 · Question 1 part b · 3 marks

2023 JAN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

Question number					
1	(a)	(i)			3
		(ii)	• support the body; • protection (of the brain and vital organs);		2
	(b)		Three from: • decrease in bone mass/bones less dense/more spongy/porous; • bone strength weakens/brittle bones/bones break more easily/fractures occur; • height decreases/stoop/poor posture; • pain;		3

Total 8 marks

Type MS · Explain dietary needs for bone and muscle and describe osteoporosis

Answer 5 of 5 · Question 1 part b · 3 marks

2023 JAN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks